

The rotary switch-selectable timer for industrial and domestic applications is designed as a robust and practical timing device for systems that require fixed delays with minimal user intervention. In many real-world settings, particularly on shop floors and within household control units, simple hardware-based timers are often preferred over programmable devices. A rotary switch position enables quick selection of the required delay time, allowing even non-technical persons to operate an appliance easily.

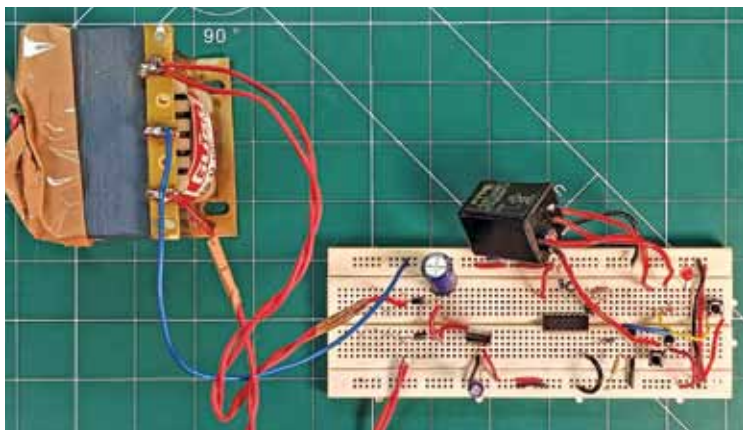
In industrial environments, such timers are commonly used for motor start delay, conveyor sequencing, lubrication systems, heater control, and protection circuits where a defined waiting period is required before switching a load. In domestic settings, the device supports applications such as pump automation, staircase or garden lighting, exhaust fans, water heaters, and appliance

protection. The ability to select preset timings improves operational consistency and reduces the risk of accidental misconfiguration.

The design does not rely on software. By avoiding microcontrollers, the circuit eliminates concerns such as program corruption, debugging complexity, and dependence on specialised tools. As a result, the device remains well suited for harsh environments, long-term operation, and straightforward maintenance. Fig. 1 shows the author's prototype.

Circuit and working

Fig. 2 shows the circuit diagram of the rotary switch-selectable timer for industrial and domestic use. The device is built around



The circuit diagram illustrates a 12V SPDT relay control system. The power supply consists of a 230V AC source connected through a fuse F1 (0.5A) to a transformer X1. The secondary winding of X1 provides 12V-0-12V AC, which is rectified by diodes D1 and D2 (1N4007) and filtered by capacitor C1 (1000µF, 25V). A voltage divider consisting of R1 (1K) and R4 (2.2K) is used to provide a reference voltage for LED1 (ON). The main control logic is based on IC2, a CD4060BE counter, which is configured with VDD and VSS pins. The counter's output Q13 is connected to the base of transistor T1 (BC548), which drives the coil of the relay. The relay's common terminal is connected to the 12V supply, and its two switchable terminals are connected to the load (APPLIANCE) and ground. The relay is also controlled by a 12V SPDT switch S1 (START) and a stop button S2. A 1N4007 diode D3 is connected across the relay coil to protect it from back EMF. Other components include resistors R2 (2M), R3 (120K), R5 (1K), R6 (10K), and capacitors C2 (1µF, 25V) and C3 (4.7µF, 25V).

Fig. 2: Circuit diagram